

Fig. 4: Performance scores (in seconds) of different processors on two speed tests

microarchitecture processor working on your computer, you will also need to purchase a motherboard supporting the LGA 1150 socket.

Last year Intel's president and CEO, Paul Otellini, revealed how advances in silicon technology and platform engineering will reduce idle platform power for the Haswell chip in 2013 by more than 20 times over current technology designs. He also demonstrated a PC with its processor running on solar power.

Intel Haswell processors started showing up for platform verification in mid-January, 2012. Since Ivy Bridge processors had arrived at the same time last year, Haswell could be released well before the second quarter of 2013.

Lynx Point, the platform controller hub on the Intel Haswell processors, is also reported to be visibly smaller. This speaks a lot about the on-die integration done by Intel.

Comparison of popular CPUs

Fig. 4 shows performance scores using two speed test tools, the Adobe PS CS4 Retouch Artists and the Windows Media Encoder 9. The Adobe Photoshop CS4 Retouch Artists is a multithreaded test and thus it can utilise all the cores available in the machine. Similarly, the video encoding test also utilises all the cores effectively.

This chart shows how different processors perform on two speed tests. The processors compared are Intel's first-generation Nehalem-based Core-i processors, AMD Bulldozer based Opteron processor and Intel's second-generation Sandy Bridge based processor. As you can see, the new architecture of the Sandy Bridge processor gives it a significant boost in performance against its older Nehalem competitors. Even the two-billion-transistor Bulldozer processor from AMD struggles

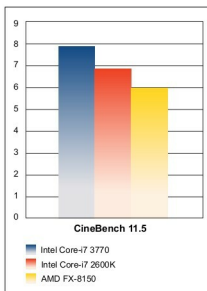


Fig. 5: Comparison of the predicted Ivy Bridge value with known values of the Sandy Bridge and AMD Bulldozer processors using CineBench 11.5 tool

to compete.

Fig. 5 utilises Intel's performance expectation values for the Ivy Bridge processor to create a chart that compares the predicted Ivy Bridge value with known values of the Sandy Bridge and AMD Bulldozer processors.

In a nutshell

Now that you know how the different Intel Core-i series processors stand up to each other and to the AMD Bulldozer microarchitecture, you can decide better what microprocessor to look for in your next gaming rig or high-performance PC. You can also decide whether to go for Ivy Bridge or just a current Sandy Bridge or Bulldozer. Or perhaps, you could think long-term and save up for a Haswell that is due next year. ●

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